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## Data and Applications

Instructor : #PKGiri

→ Data :

Known facts that can be recorded and have an implicit meaning

Database :

Collection of related data

DBMS :

DB + Software

Mini-world :

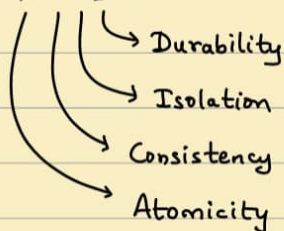
Some part of the real world about which data is stored in DB.

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Triggers and Stored Procedure → Not part of syllabus

Keywords : Entity , Attribute , Relations

- ACID



- Constraints , Operations , Datatypes

Database schema → Description | Catalogue

State → Snapshot

- Three - State Architecture

Two - tier Client - Server architecture

- DDL ← Data creation

DML ← Query Language

→ Create the following for an event of Infinium :

- (a) Tables
  - (b) Columns
  - (c) Datatypes
  - (d) Relationships
- } Database Schema

### Database schema :

Infinium → Mini-world

e.g. Euphony :

STUDENTS form a band with other STUDENTS.

BAND has multiple roles - Singer, Pianist, Guitarist, Drummer, etc.

BAND gives auditions and proceed further if selected.

CLUB MEMBERS judge the auditions. They are also STUDENTS.

BANDS performs SONGs.

BANDS are given positions based on their performance

| Band ID | Performer ID |
|---------|--------------|
| int     | int          |

| Performer ID | Talent  |
|--------------|---------|
| int          | varchar |

| Band ID | Selected/NOT |
|---------|--------------|
| int     | varchar      |

| Band ID | Song ID |
|---------|---------|
| int     | int     |

| Band ID | Rank |
|---------|------|
| int     | int  |

| Song ID | Song name |
|---------|-----------|
| int     | varchar   |

1 Band → n Performers

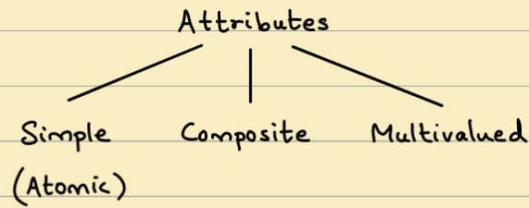
1 Band → n songs

1 Performer → 1 Talent

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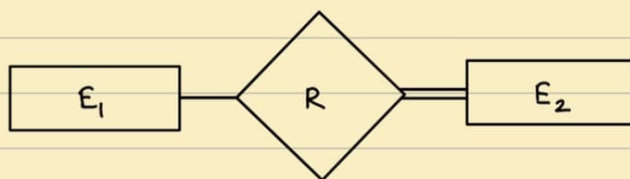
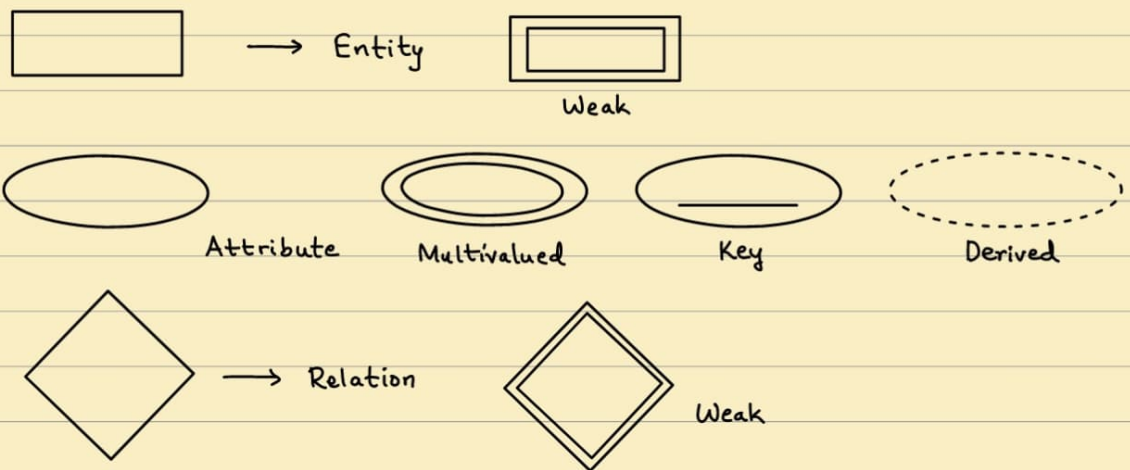
## → ER Model

- Database Designer : doesn't care how the data is captured.
- Entity : Basic concept of ER Model (Row)
- Attributes : Properties of entity (column)

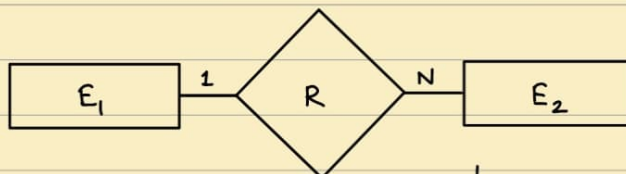


Key Attribute → Each entity has a unique value

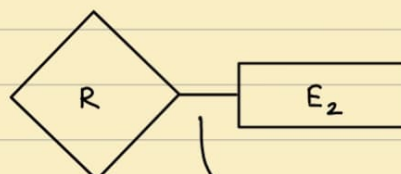
Representation : Entity. Attribute



→ Total participation of  $E_2$  in  $R$



→ Cardinality Ratio → 1:N for  $E_1:E_2$  in  $R$



→ (min-max) → structural constraint

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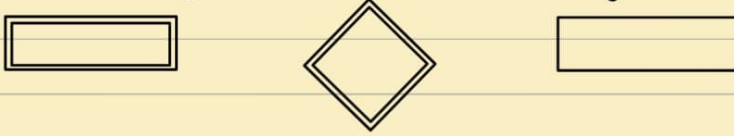
Relation types

Constraints

Recursive relations

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- Weak Entity  $\rightarrow$  Depends on strong Entity



- Attributes of Relationship types

- Cardinality ratio

Participation constraint

- min - max notation

If you query often  $\rightarrow$  Derived Attribute X  
Relationship type  $\checkmark$

$\therefore$  Fetching is much faster

$\rightarrow$  Data Models:

(1) ER Models

(2) UML Class diagram

Entities  $\leftrightarrow$  Classes

Relationships  $\leftrightarrow$  Lines b/w classes

$\rightarrow$  DBMS:

Scripting Language

Embedded Approach

- Interface

- Utilities  $\rightarrow$  Functions

- Cambridge Analytica - 3500 features stored about you

- Centralised DBMS Architecture

Client - Server Architecture  $\rightarrow$  3-Tier

## → Relational Model :

- Relation → Set of rows, i.e. tuple  
Attribute → Column  
→ Ordered set of values (1 tuple : 1 Row)

- Key : Uniquely identifies a row in a table  
→ Can be set of attributes (or) attribute

- Schema : Table definition

$R(A_1, A_2, \dots, A_n)$   
→ Attributes  
→ name of the relation

- Domain :

- All possible column values
- Datatype constraint  
Format constraint
- Developed from Attribute name

- Relation State :

$r(R)$  → Subset of Relation  $R$

$$r(R) = \{t_1, t_2, \dots, t_n\}$$

$$t_i = \langle v_1, v_2, \dots, v_n \rangle$$

$v_j$  → element of domain of  $A_j$

$$r(R) \subset \text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n)$$

The tuples are not considered to be ordered.

The attributes are ordered. ← Self-Describing